

Asymptotic and structural results for OEIS sequence A112509

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Abstract

For a binary string s of length n , let $f(s)$ denote the number of distinct non-negative integers whose binary representations appear as contiguous substrings of s (leading zeros suppressed), and set $a(n) = \max_{|s|=n} f(s)$ (OEIS A112509 [1]). We prove three results. First, $\lim_{n \rightarrow \infty} a(n)/n^2 = \frac{1}{2}$; more precisely, for all $n \geq 64$,

$$\frac{1}{2}n^2 - 15n^{3/2} \leq a(n) \leq \frac{n(n+1)}{2},$$

so $a(n)/n^2 = \frac{1}{2} + O(n^{-1/2})$. Second, every optimal string has ones-density $1 - O(n^{-1/4})$; explicit near-optimal strings achieving this bound are constructed. Third, $a(n)$ is convex: $a(n+1) - 2a(n) + a(n-1) \geq 0$ for all $n \geq 2$, so the first differences $\Delta a(n)$ are nondecreasing.

1 Introduction

1.1 The sequence

For $s \in \{0,1\}^n$, let $f(s)$ denote the number of distinct non-negative integers whose binary representations appear as contiguous substrings of s (leading zeros suppressed: $01 = 1$, $00 = 0$). Set $a(n) = \max_{|s|=n} f(s)$. This is OEIS A112509 [1], beginning

$$1, 3, 5, 7, 10, 13, 17, 22, 27, 33, 40, 47, 55, 64, 73, 83, 94, 106, 118, 131, \dots$$

1.2 Numerical evidence

Exact values for $n \leq 80$ are due to M. Fuller and appear in the OEIS b-file [1]; no algorithm or proof of completeness was published. Heuristic values for $81 \leq n \leq 150$ were computed in [2] using multiple independent Metropolis–Hastings and template-search methods, with consistent agreement between methods but no formal proof of optimality. Certified lower bounds for much larger n were obtained via suffix-array-guided local search [2]. Table 1 records selected values.

n	$a(n)$ or lower bound	$a(n)/n^2 \geq$
50	899	0.3596
100	3,875	0.3875
150	9,070	0.4031
10^6	498,396,662,200	0.4984
10^9	499,943,911,277,037,650	0.49994
2×10^9	1,999,841,302,297,432,600	0.49996

Table 1: Selected values and lower bounds for $a(n)$. Values for $n \leq 80$ are from Fuller [1]; values for $81 \leq n \leq 150$ are heuristic computational estimates [2]; large- n entries are certified lower bounds [2].

1.3 Main results

Theorem 1 (Limit). $\lim_{n \rightarrow \infty} \frac{a(n)}{n^2} = \frac{1}{2}$.

Theorem 2 (Quantitative lower bound). *For every $n \geq 64$,*

$$a(n) \geq \frac{1}{2}n^2 - 15n^{3/2}.$$

Proposition 3 (Density of ones).

- (i) *There exists an explicit family of strings s_n with $f(s_n) \geq \frac{1}{2}n^2 - 15n^{3/2}$ and $N_0(s_n) = O(\sqrt{n})$, so the ones-density satisfies $N_1(s_n)/n = 1 - O(n^{-1/2})$.*
- (ii) *Every optimal string s_n^* satisfies $N_0(s_n^*) = O(n^{3/4})$, equivalently $N_1(s_n^*)/n = 1 - O(n^{-1/4})$.*

Theorem 4 (Convexity). *For every $n \geq 2$,*

$$a(n+1) - 2a(n) + a(n-1) \geq 0.$$

Equivalently, $\Delta a(n) := a(n) - a(n-1)$ is nondecreasing for $n \geq 2$.

Section 3 gives the upper bound. Sections 4–6 prove Theorems 1–2. Section 8 proves Proposition 3. Sections 9–10 develop the structural bounds. Section 11 proves Theorem 4.

2 Definitions

If $w = w_1 \cdots w_k \in \{0, 1\}^k$, its binary value is $\nu(w) = \sum_{i=1}^k w_i 2^{k-i}$. The value set of $s \in \{0, 1\}^n$ is $V(s) = \{\nu(s_i \cdots s_j) : 1 \leq i \leq j \leq n\}$, and $f(s) = |V(s)|$. Write $N_1(s)$ and $N_0(s)$ for the number of ones and zeros in s .

3 Upper bound

Lemma 5 (Upper bound). *For every $n \geq 1$, $a(n) \leq n(n+1)/2$, and hence $\limsup_{n \rightarrow \infty} a(n)/n^2 \leq 1/2$.*

Proof. Every distinct positive integer in $V(s)$ has a unique standard binary representation beginning with 1, so it is represented by at least one substring of s beginning with 1. Different positive integers require different such substrings.

If s contains at least one zero then $0 \in V(s)$, represented only by substrings beginning with 0. At least one of the $n(n+1)/2$ substrings begins with 0, so the number of leading-1 substrings is at most $n(n+1)/2 - 1$, giving $f(s) \leq n(n+1)/2$. If s is all ones then $f(s)$ counts distinct positive values, at most $n(n+1)/2$. $\square$

Remark 6. The bound is tight at $n = 2$: 10 gives $V(10) = \{0, 1, 2\}$, so $a(2) = 3 = 2 \cdot 3/2$.

4 The construction

4.1 Strictly decreasing partitions

Lemma 7 (Decreasing partition). *Let $k \geq 2$ and $M \geq k(k+1)/2$. Then there exist strictly decreasing positive integers $b_1 > b_2 > \cdots > b_k \geq 1$ with $\sum_{i=1}^k b_i = M$. Moreover, they may be chosen with $|b_i - M/k| \leq (k+1)/2$ for all i .*

Proof. Strategy. Consider the arithmetic sequence $c_i = q + k - i$ ($i = 1, \dots, k$), which decreases by 1 at each step. Its sum is

$$\sum_{i=1}^k c_i = kq + \sum_{i=1}^k (k - i) = kq + \frac{k(k-1)}{2}. \quad (1)$$

Choose q so this base sum is as large as possible without exceeding M , and let $r = M - \sum c_i$ be the shortfall. We then add 1 to the first r terms to account for the shortfall.

Definitions. Set

$$q = \left\lfloor \frac{M}{k} - \frac{k-1}{2} \right\rfloor, \quad r = M - kq - \frac{k(k-1)}{2},$$

and define

$$b_i = \begin{cases} q + k - i + 1, & 1 \leq i \leq r, \\ q + k - i, & r < i \leq k. \end{cases}$$

Step 1: $0 \leq r < k$. From the definition of q as a floor, $q \leq M/k - (k-1)/2 < q + 1$. Multiplying through by k gives

$$kq \leq M - \frac{k(k-1)}{2} < k(q+1),$$

so $0 \leq M - kq - k(k-1)/2 < k$, i.e. $0 \leq r < k$.

Step 2: Strict decrease. We check all three cases for consecutive pairs $b_i > b_{i+1}$.

- *Both in the first case* ($i < r$, so $i+1 \leq r$ as well): $b_i - b_{i+1} = (q+k-i+1) - (q+k-(i+1)+1) = 1 > 0$.
- *Transition* ($i = r \geq 1$, so $i+1 > r$): $b_r = q + k - r + 1$ and $b_{r+1} = q + k - (r+1)$, giving $b_r - b_{r+1} = q + k - r + 1 - (q + k - r - 1) = 2 > 0$.
- *Both in the second case* ($i > r$): $b_i - b_{i+1} = (q + k - i) - (q + k - i - 1) = 1 > 0$.

Hence $b_1 > b_2 > \dots > b_k$.

Step 3: The sum equals M . Split the sum at the boundary between the two cases:

$$\begin{aligned} \sum_{i=1}^k b_i &= \sum_{i=1}^r (q + k - i + 1) + \sum_{i=r+1}^k (q + k - i) \\ &= \sum_{i=1}^r (q + k - i) + \sum_{i=1}^r 1 \\ &= kq + \frac{k(k-1)}{2} + r = M, \end{aligned}$$

where the last equality uses the definition of r and (1).

Step 4: All terms are positive. From $M \geq k(k+1)/2$ we get $M/k \geq (k+1)/2$, so

$$\frac{M}{k} - \frac{k-1}{2} \geq \frac{k+1}{2} - \frac{k-1}{2} = 1,$$

hence $q = \lfloor M/k - (k-1)/2 \rfloor \geq 1$. Since $r < k$, the index $i = k$ falls in the second case, giving $b_k = q + k - k = q \geq 1$.

Step 5: Proximity bound. Every first-case term satisfies $q < q + k - r + 1 \leq b_i \leq q + k$, and every second-case term satisfies $q \leq b_i = q + k - i \leq q + k - r - 1 < q + k$. So all $b_i \in [q, q + k]$. From the floor definition, $M/k \in [q + (k-1)/2, q + (k+1)/2)$, and the width of $[q, q + k]$ around any point in this interval is at most $(k+1)/2$ in each direction. Hence $|b_i - M/k| \leq (k+1)/2$ for all i . $\square$

4.2 Parameters and string

Fix $n \geq 64$ and set

$$k = \left\lfloor \frac{\sqrt{n}}{4} \right\rfloor, \quad M = n - (k - 1). \quad (2)$$

Since $n \geq 64$ we have $k = \lfloor \sqrt{n}/4 \rfloor \geq \lfloor 8/4 \rfloor = 2$.

Why $M \geq n/2$. Since $k \leq \sqrt{n}/4$, we have $\sqrt{n}/4 \leq n/2$ (equivalently $\sqrt{n} \leq 2n$, true for all $n \geq 1$), so $k - 1 < k \leq \sqrt{n}/4 \leq n/2$. Hence $M = n - (k - 1) > n - n/2 = n/2$.

Why $n/2 \geq k(k + 1)/2$. This is equivalent to $n \geq k(k + 1)$. Since $k \leq \sqrt{n}/4$ we have

$$k(k + 1) \leq k \cdot 2k = 2k^2 \leq 2 \cdot \frac{n}{16} = \frac{n}{8} \leq n,$$

where the second inequality uses $k \leq \sqrt{n}/4$ so $k^2 \leq n/16$. Hence $M \geq n/2 \geq k(k + 1)/2$, and Lemma 7 applies.

Choose $b_1 > b_2 > \dots > b_k \geq 1$ with $\sum b_i = M$ and $|b_i - M/k| \leq k$, as in Lemma 7. Define

$$s = \underbrace{1^{b_1}}_{\text{block 1}} 0 \underbrace{1^{b_2}}_{\text{block 2}} 0 \dots 0 \underbrace{1^{b_k}}_{\text{block } k}, \quad (3)$$

consisting of k one-blocks separated by $k - 1$ single zeros. Its length is $\sum b_i + (k - 1) = M + (k - 1) = n$.

5 An injective family of substrings

For integers i, j, α, β satisfying

$$1 \leq i, \quad j \geq i + 2, \quad j \leq k, \quad 1 \leq \alpha \leq b_i, \quad 1 \leq \beta \leq b_j, \quad (4)$$

define $u_{i,j,\alpha,\beta}$ to be the substring of s beginning α positions before the end of block i and ending after the first β ones of block j . In binary,

$$u_{i,j,\alpha,\beta} = 1^\alpha 0 1^{b_{i+1}} 0 1^{b_{i+2}} 0 \dots 1^{b_{j-1}} 0 1^\beta. \quad (5)$$

The condition $j \geq i + 2$ ensures at least one full interior block $(b_{i+1}, \dots, b_{j-1})$. Every $u_{i,j,\alpha,\beta}$ starts and ends with 1, so it is already a standard binary expansion of a positive integer.

Lemma 8 (Injectivity). *The map $(i, j, \alpha, \beta) \mapsto u_{i,j,\alpha,\beta}$ is injective on the range (4).*

Proof. Suppose $u_{i,j,\alpha,\beta} = u_{i',j',\alpha',\beta'}$ as binary words. The run-length encoding (RLE) of (5), listing alternating run-lengths of ones then zeros, is

$$(\alpha, 1, b_{i+1}, 1, b_{i+2}, 1, \dots, b_{j-1}, 1, \beta).$$

Matching entries gives $\alpha = \alpha'$, $\beta = \beta'$, and $(b_{i+1}, \dots, b_{j-1}) = (b_{i'+1}, \dots, b_{j'-1})$. Since $b_1 > \dots > b_k$ is strictly decreasing, every consecutive subsequence occurs in exactly one position, so $i = i'$ and $j = j'$. $\square$

6 Quantitative lower bound and the limit

Proof of Theorem 2. By Lemma 8, the family $\{u_{i,j,\alpha,\beta}\}$ consists of pairwise distinct positive integers, so

$$a(n) \geq f(s) \geq \sum_{\substack{1 \leq i < j \leq k \\ j \geq i+2}} b_i b_j. \quad (6)$$

Expanding:

$$\sum_{j \geq i+2} b_i b_j = \frac{(\sum b_i)^2 - \sum b_i^2}{2} - \sum_{i=1}^{k-1} b_i b_{i+1}. \quad (7)$$

Step 1. Since $\sum b_i = M = n - k + 1$ and $k \leq \sqrt{n}/4$,

$$\frac{M^2}{2} \geq \frac{(n - \sqrt{n}/4)^2}{2} \geq \frac{n^2}{2} - \frac{n^{3/2}}{4}.$$

Step 2. Write $b_i = M/k + d_i$ with $|d_i| \leq k$. Then $\sum b_i^2 \leq M^2/k + k^3$. Since $k \geq \sqrt{n}/8$ (as $\lfloor x \rfloor \geq x/2$ for $x \geq 2$), $M^2/k \leq n^2/(\sqrt{n}/8) = 8n^{3/2}$, and $k^3 \leq (\sqrt{n}/4)^3 = n^{3/2}/64$, so

$$\sum_{i=1}^k b_i^2 < 9n^{3/2}. \quad (8)$$

Step 3. By AM–GM, $b_i b_{i+1} \leq (b_i^2 + b_{i+1}^2)/2$, so

$$\sum_{i=1}^{k-1} b_i b_{i+1} \leq \sum_{i=1}^k b_i^2 < 9n^{3/2}. \quad (9)$$

Conclusion. Substituting into (7):

$$a(n) \geq \frac{n^2}{2} - \frac{n^{3/2}}{4} - \frac{9n^{3/2}}{2} - 9n^{3/2} = \frac{n^2}{2} - \frac{55}{4}n^{3/2} > \frac{n^2}{2} - 15n^{3/2}. \quad \square$$

Proof of Theorem 1. Lemma 5 gives $\limsup a(n)/n^2 \leq 1/2$. Theorem 2 gives $a(n)/n^2 \geq 1/2 - 15/\sqrt{n} \rightarrow 1/2$. $\square$

Corollary 9. For all $n \geq 64$, $n^2/2 - 15n^{3/2} \leq a(n) \leq n(n+1)/2$, so $a(n)/n^2 = 1/2 + O(n^{-1/2})$.

Remark 10. The lower bound $n^2/2 - 15n^{3/2}$ is negative for $n < 900$ (since $n^2/2 > 15n^{3/2}$ iff $\sqrt{n} > 30$). For such n the inequality in Theorem 2 is trivially satisfied by the non-negativity of $a(n)$, and carries no information. The bound is meaningful only for $n \geq 900$; for small n the exact values from computational search (Table 1) give much sharper information.

6.1 Sharper quantitative lower bound

The constant 15 in Theorem 2 is an artefact of the parameter choice $k = \lfloor \sqrt{n}/4 \rfloor$. Choosing instead $k = \lceil \sqrt{n} \rceil$, together with a slightly sharper bound on the variance of the partition produced by Lemma 7, reduces the constant by a factor of $60/13 > 4.6$, and makes the bound non-trivial already at $n = 36$ rather than $n = 900$.

Theorem 11 (Sharper quantitative lower bound). *For every $n \geq 36$,*

$$a(n) \geq \frac{n^2}{2} - \frac{13}{4}n^{3/2}.$$

Remark 12. Comparison with Theorem 2:

- The new bound is positive iff $\sqrt{n} > 13/2$, i.e. $n \geq 43$, while the old bound is positive only for $n \geq 901$. Thus the threshold beyond which the bound is informative drops by a factor of more than 20.
- The constant in front of $n^{3/2}$ improves from 15 to $13/4$, a factor of $60/13 > 4.6$. In fact the construction achieves a constant of $\sqrt{6} + o(1) \approx 2.449$ (Remark 13 below); we keep $13/4$ here because it admits a fully symbolic and elementary proof.

A numerical comparison at sample n (with $a(n)$ taken from the OEIS b-file b112509):

n	Thm. 2	Thm. 11	$a(n)$
36	-2790	-54	450
50	-3050	101	899
100	-10000	1750	3875
150	-16320	5278	9070
1000	-237201	397214	—

Proof of Theorem 11. The construction (3) and Lemmas 7 and 8 are unchanged. Only the parameter choice and the estimation of the loss terms in (7) differ.

Parameter. Set $k = \lceil \sqrt{n} \rceil$ and $M = n - k + 1$. Then $\sqrt{n} \leq k \leq \sqrt{n} + 1$.

Lemma 7 applies for $n \geq 36$. We need $k \geq 2$ (clear) and $M \geq k(k+1)/2$. The latter is equivalent to $n - \sqrt{n} \geq \frac{1}{2}(\sqrt{n} + 1)(\sqrt{n} + 2) = \frac{n}{2} + \frac{3\sqrt{n}}{2} + 1$ (using $k \leq \sqrt{n} + 1$), i.e. $\frac{n}{2} \geq \frac{5\sqrt{n}}{2} + 1$, which holds for $\sqrt{n} \geq 6$, i.e. $n \geq 36$.

Variance bound. By Lemma 7, every b_i lies in $[q, q+k]$ where $q = \lfloor M/k \rfloor$. Since $\sum_{i=1}^k (b_i - M/k) = 0$ and each deviation has absolute value at most $k/2 + 1 \leq (k+1)/2$ (because $b_i \in \{q, \dots, q+k\}$ and the average is $M/k \in [q, q+k]$),

$$\sum_{i=1}^k b_i^2 = \frac{M^2}{k} + \sum_{i=1}^k \left(b_i - \frac{M}{k}\right)^2 \leq \frac{M^2}{k} + k \cdot \frac{(k+1)^2}{4}. \quad (10)$$

By the AM–GM inequality $2b_i b_{i+1} \leq b_i^2 + b_{i+1}^2$,

$$\sum_{i=1}^{k-1} b_i b_{i+1} \leq \sum_{i=1}^k b_i^2 \leq \frac{M^2}{k} + \frac{k(k+1)^2}{4}. \quad (11)$$

Substituting (10) and (11) into (7):

$$\begin{aligned} a(n) &\geq \frac{M^2 - \sum b_i^2}{2} - \sum_{i=1}^{k-1} b_i b_{i+1} \\ &\geq \frac{M^2}{2} - \frac{1}{2} \left(\frac{M^2}{k} + \frac{k(k+1)^2}{4} \right) - \left(\frac{M^2}{k} + \frac{k(k+1)^2}{4} \right) \\ &= \frac{M^2}{2} - \frac{3M^2}{2k} - \frac{3k(k+1)^2}{8}. \end{aligned}$$

Term-by-term estimates. Using $\sqrt{n} \leq k \leq \sqrt{n} + 1$ and $M = n - k + 1$:

- (i) $\frac{M^2}{2} \geq \frac{(n - \sqrt{n})^2}{2} = \frac{n^2}{2} - n^{3/2} + \frac{n}{2}$.
- (ii) $\frac{3M^2}{2k} \leq \frac{3n^2}{2\sqrt{n}} = \frac{3}{2}n^{3/2}$.
- (iii) Expanding $(\sqrt{n} + 1)(\sqrt{n} + 2)^2 = n^{3/2} + 5n + 8\sqrt{n} + 4$ gives

$$\frac{3k(k+1)^2}{8} \leq \frac{3(\sqrt{n} + 1)(\sqrt{n} + 2)^2}{8} = \frac{3}{8}n^{3/2} + \frac{15}{8}n + 3\sqrt{n} + \frac{3}{2}.$$

Combining (i)–(iii):

$$\begin{aligned} a(n) &\geq \frac{n^2}{2} - n^{3/2} + \frac{n}{2} - \frac{3}{2}n^{3/2} - \frac{3}{8}n^{3/2} - \frac{15}{8}n - 3\sqrt{n} - \frac{3}{2} \\ &= \frac{n^2}{2} - \frac{23}{8}n^{3/2} - \frac{11}{8}n - 3\sqrt{n} - \frac{3}{2}. \end{aligned} \quad (12)$$

Tail bound. We claim

$$\frac{11}{8}n + 3\sqrt{n} + \frac{3}{2} \leq \frac{3}{8}n^{3/2} \quad (n \geq 36). \quad (13)$$

Equivalently $11n + 24\sqrt{n} + 12 \leq 3n^{3/2}$. At $n = 36$ the right side equals $3 \cdot 216 = 648$ and the left side equals $396 + 144 + 12 = 552$, so the inequality holds at $n = 36$ with margin 96. Differentiating, $\frac{d}{dn}(3n^{3/2} - 11n - 24\sqrt{n} - 12) = \frac{9}{2}\sqrt{n} - 11 - \frac{12}{\sqrt{n}}$, which is positive for $n \geq 36$ since $\frac{9}{2} \cdot 6 - 11 - 2 = 14 > 0$ and the derivative is monotone increasing. Therefore the inequality holds for all $n \geq 36$.

Combining (12) and (13),

$$a(n) \geq \frac{n^2}{2} - \frac{23}{8}n^{3/2} - \frac{3}{8}n^{3/2} = \frac{n^2}{2} - \frac{26}{8}n^{3/2} = \frac{n^2}{2} - \frac{13}{4}n^{3/2}. \quad \square$$

Remark 13. The constant $13/4$ is not optimal. Tracking the variance of the partition more carefully (it is bounded by $k(k^2-1)/12 + O(k)$ rather than $k(k+1)^2/4$) and choosing $k = \lceil \sqrt{3n/2} \rceil$ instead of $\lceil \sqrt{n} \rceil$ yields

$$a(n) \geq \frac{n^2}{2} - \sqrt{6}n^{3/2} + O(n),$$

with $\sqrt{6} \approx 2.449$. The choice $k = \lceil \sqrt{3n/2} \rceil$ is the one that equates the two leading $n^{3/2}$ losses kn and $3n^2/(2k)$ in the proof above. We do not record the technical details since they would obscure the elementary nature of Theorem 11.

Corollary 14. *For all $n \geq 36$, $a(n)/n^2 = \frac{1}{2} + O(n^{-1/2})$ with explicit constants: $\frac{1}{2} - \frac{13}{4}n^{-1/2} \leq a(n)/n^2 \leq \frac{1}{2} + \frac{1}{2n}$.*

7 A structural constraint on optima

In this section we prove a rigorous upper bound on $f(s)$ in terms of the run-length statistics of s . The bound is universal: it holds for *every* binary string and is tight enough to force quantitative structural constraints on every optimum of $a(n)$.

7.1 Run-length notation

For $s \in \{0, 1\}^n$, write its *run decomposition*

$$s = a^{r_1} \bar{a}^{r_2} a^{r_3} \dots$$

where $a \in \{0, 1\}$ is the leading symbol and the maximal runs have lengths $r_1, r_2, \dots$ alternating between a and $\bar{a}$. Let $\ell_1, \dots, \ell_p$ denote the lengths of the 1-runs and $m_1, \dots, m_q$ the lengths of the 0-runs (so $p + q \leq n + 1$ and $\sum \ell_t + \sum m_t = n$). Set

$$L(s) = \max_t \ell_t, \quad K(s) = \max_t m_t,$$

the *longest 1-run* and *longest 0-run* respectively (with the convention $L(s) = 0$ if s contains no 1s, and $K(s) = 0$ if s contains no 0s).

7.2 The run-length frequency bound

We work with two versions of the bound: a tighter *asymmetric* form involving only the longest 1-run, and a slightly looser *symmetric* form that lets us simultaneously control both the longest 1-run and the longest 0-run.

Theorem 15 (Run-length frequency bound, asymmetric). *Every $s \in \{0, 1\}^n$ satisfies*

$$f(s) \leq 1 + L(s) + \binom{n+1}{2} - \sum_{t=1}^p \binom{\ell_t+1}{2} - \sum_{t=1}^q \binom{m_t+1}{2}. \quad (14)$$

Proof. Recall $f(s) = |V(s)|$, where $V(s) \subseteq \mathbb{Z}_{\geq 0}$ is the set of integer values represented by substrings of s , with the empty substring contributing the value 0. Partition

$$V(s) = \{0\} \cup V_1(s) \cup V_m(s), \quad (15)$$

where $V_1(s)$ is the set of values represented by some all-1 substring of s (necessarily $\neq 0$), and $V_m(s) := V(s) \setminus (\{0\} \cup V_1(s))$ collects the remaining values. This is a disjoint union: $0 \notin V_1$ since every length- ≥ 1 all-1 substring has value $2^j - 1 \geq 1$, and V_m is defined to be disjoint from the other two sets.

Counting $V_1(s)$. A nonempty substring of the form $0^a 1^j$ ($a \geq 0, j \geq 1$) has integer value $2^j - 1$, depending only on j . Hence $V_1(s) = \{2^j - 1 : 1 \leq j \leq L(s)\}$ and $|V_1(s)| = L(s)$.

Counting $V_m(s)$. Every value in $V_m(s)$ is represented by at least one substring, and that substring must contain both a 0 and a 1: otherwise it would be a power of 2 minus one (hence in V_1) or zero. Therefore $|V_m(s)|$ is bounded above by the number of pairs (i, j) with $1 \leq i \leq j \leq n$ such that $s_i \cdots s_j$ contains both symbols. This count equals $\binom{n+1}{2}$ minus the number of (i, j) for which $s_i \cdots s_j$ is all-1 or all-0. A pair (i, j) yields an all-1 substring iff $\{i, \dots, j\}$ is contained in some maximal 1-run, and the number of such pairs within a run of length ℓ is $\binom{\ell+1}{2}$. Summing over 1-runs and 0-runs,

$$|V_m(s)| \leq \binom{n+1}{2} - \sum_{t=1}^p \binom{\ell_t+1}{2} - \sum_{t=1}^q \binom{m_t+1}{2}.$$

Combining with $|V_1(s)| = L(s)$ via (15) gives (14). $\square$

Remark 16. Inequality (14) has been verified to hold with strictly positive slack for every optimal string in our database ($1 \leq n \leq 150$, totalling 21 072 optima). At sample n the slack $\text{RHS} - a(n)$ behaves like:

n	10	30	50	80	100	130	150
slack	7	77	176	436	614	965	1 241

matching the predicted $\Theta(n^{3/2})$ scaling of the gap $\binom{n+1}{2} - a(n)$. The bound becomes equality when every pair (i, j) yielding a mixed substring gives an integer value not already counted; this fails for long s because of the leading-zero collision $0^a 1^j 0^b \sim 1^j 0^b$.

7.3 Forced run-length statistics at any optimum

Corollary 17 (Concentration of run lengths). *Let s be any optimum for $a(n)$ with run lengths $\ell_1, \dots, \ell_p, m_1, \dots, m_q$ as above and longest 1-run $L = L(s)$. Then for all $n \geq 36$,*

$$\sum_{t=1}^p \binom{\ell_t+1}{2} + \sum_{t=1}^q \binom{m_t+1}{2} \leq \frac{13}{4} n^{3/2} + \frac{n}{2} + L + 1. \quad (16)$$

Proof. Apply Theorem 15 with $f(s) = a(n)$ and rearrange:

$$\sum_t \binom{\ell_t+1}{2} + \sum_t \binom{m_t+1}{2} \leq \binom{n+1}{2} + 1 + L - a(n).$$

Using $\binom{n+1}{2} = \frac{n^2}{2} + \frac{n}{2}$ and the lower bound $a(n) \geq \frac{n^2}{2} - \frac{13}{4} n^{3/2}$ of Theorem 11, $\binom{n+1}{2} - a(n) \leq \frac{n}{2} + \frac{13}{4} n^{3/2}$, giving (16). $\square$

Corollary 18 (Forced upper bound on the longest 1-run). *For every $n \geq 36$ and every optimum s for $a(n)$,*

$$\binom{L(s)+1}{2} \leq \frac{13}{4} n^{3/2} + \frac{n}{2} + L(s) + 1, \quad (17)$$

hence $L(s) \leq \sqrt{\frac{13}{2}} n^{3/4} + O(n^{3/8}) = O(n^{3/4})$.

Proof. The longest 1-run contributes $\binom{L+1}{2}$ to the left-hand side of (16); all other terms are non-negative, giving (17). Since $\binom{L+1}{2} \geq L^2/2$, the inequality forces $L^2/2 \leq \frac{13}{4}n^{3/2} + L + \frac{n}{2} + 1$. Solving and absorbing lower-order terms, $L \leq \sqrt{13/2} n^{3/4} + O(n^{3/8})$. $\square$

7.4 A symmetric bound and the longest 0-run

The argument above privileges 1-substrings because the empty substring shares the value 0 with every all-0 substring, making the all-0 substrings “invisible” to f . By treating 0-substrings on equal footing one obtains a slightly looser bound that controls both the longest 1-run and the longest 0-run.

Theorem 19 (Run-length frequency bound, symmetric). *Every $s \in \{0,1\}^n$ satisfies*

$$f(s) \leq 1 + L(s) + K(s) + \binom{n+1}{2} - \sum_{t=1}^p \binom{\ell_t+1}{2} - \sum_{t=1}^q \binom{m_t+1}{2}. \quad (18)$$

Proof. Inequality (18) follows immediately from Theorem 15 since $K(s) \geq 0$. $\square$

Corollary 20 (Forced upper bound on the longest 0-run). *For every $n \geq 36$ and every optimum s for $a(n)$,*

$$\binom{K(s)+1}{2} \leq \frac{13}{4} n^{3/2} + \frac{n}{2} + L(s) + K(s) + 1,$$

hence $K(s) = O(n^{3/4})$.

Proof. Apply Theorem 19 and rearrange exactly as in the proof of Corollary 17, retaining the $\binom{K+1}{2}$ term on the left-hand side and dropping the non-negative remaining run terms. $\square$

7.5 Empirical refinement

It is tempting to conjecture that the leading 1-run length $b_1(n)$ of an optimum is monotone non-decreasing in n . This is *false*: in the database of all optima for $1 \leq n \leq 150$ we find exactly three values of n at which the minimum leading 1-run length *decreases*, namely

$$b_1(51) = 12 < b_1(50) = 13, \quad b_1(71) = 15 < b_1(70) = 16, \quad b_1(116) = 20 < b_1(115) = 21.$$

However the slightly weaker monotonicity

$$\min_{s \in S_n} b_1(s) \geq \min_{s \in S_{n-4}} b_1(s)$$

holds for every $n \leq 150$ in our database, where S_n is the set of optima for length n . We do not have a proof of this gap-4 monotonicity; the run-length frequency bound (14) only constrains $L(s) = O(n^{3/4})$ from above and is silent on monotonicity.

The construction of §4 achieves leading 1-run length $b_1 = q + k = \Theta(\sqrt{n})$, well below the information-theoretic upper bound $O(n^{3/4})$ of Corollary 18. Whether every optimum satisfies $b_1(s) = \Theta(\sqrt{n})$ (as the data suggest) is an open problem.

8 Density of ones

8.1 Near-optimal strings

Proof of Proposition 3(i). The string $s = s_n$ from (3) satisfies $f(s_n) \geq n^2/2 - 15n^{3/2}$ (this is established directly in the proof of Theorem 2, as the lower bound on $a(n)$ is obtained by exhibiting s_n and bounding $f(s_n)$ from below).

Counting the zeros. By construction, s_n consists of k one-blocks separated by $k - 1$ single zeros, so it contains exactly $k - 1$ zeros. Since $k = \lfloor \sqrt{n}/4 \rfloor \leq \sqrt{n}/4$, we have

$$N_0(s_n) = k - 1 \leq \frac{\sqrt{n}}{4} - 1 < \frac{\sqrt{n}}{4},$$

so $N_0(s_n) = O(\sqrt{n})$.

Deducing the ones-density. Since s_n has length n ,

$$N_1(s_n) = n - N_0(s_n) \geq n - \frac{\sqrt{n}}{4},$$

and therefore

$$\frac{N_1(s_n)}{n} \geq 1 - \frac{1}{4\sqrt{n}} = 1 - O(n^{-1/2}). \quad \square$$

8.2 Zero-position penalty

Lemma 21 (Zero-position bound). *Let $s \in \{0, 1\}^n$ have zeros at positions $p_1 < \dots < p_z$. Then*

$$f(s) \leq \frac{n(n+1)}{2} - \sum_{r=1}^z (n - p_r + 1).$$

In particular, $f(s) \leq n(n+1)/2 - z(z+1)/2$.

Proof. For each $x \in V(s)$ choose a minimum-length representative w_x . The map $x \mapsto w_x$ is injective. If $x > 0$ then w_x starts with 1 (otherwise removing leading zeros gives a shorter representative). So distinct positive values correspond to distinct leading-1 substrings.

First inequality. A substring starting at position p_r (which holds a zero) has the form $0s_{p_r+1} \dots s_j$ for some $j \geq p_r$, so it starts with 0. The number of substrings that start at position p_r is $n - p_r + 1$ (one for each ending position $j \in \{p_r, \dots, n\}$). None of these substrings starts with 1, so none can be the minimum-length representative w_x of any positive integer x . Removing all such substrings from consideration, the number of available starting positions for positive-value representatives is at most $n - z$ (the positions that hold a 1), and the total count of leading-1 substrings is at most $n(n+1)/2 - \sum_{r=1}^z (n - p_r + 1)$, giving the first inequality.

Second inequality. We minimise $\sum_{r=1}^z (n - p_r + 1)$ subject to $1 \leq p_1 < p_2 < \dots < p_z \leq n$. Since each term $n - p_r + 1$ is decreasing in p_r , the sum is minimised by taking the p_r as large as possible, i.e. $p_r = n - z + r$ for $r = 1, \dots, z$. Substituting:

$$\sum_{r=1}^z (n - p_r + 1) = \sum_{r=1}^z (n - (n - z + r) + 1) = \sum_{r=1}^z (z - r + 1) = \sum_{t=1}^z t = \frac{z(z+1)}{2},$$

where we set $t = z - r + 1$. Hence the minimum of $\sum (n - p_r + 1)$ is $z(z+1)/2$, so $f(s) \leq n(n+1)/2 - z(z+1)/2$. $\square$

8.3 Optimal strings have few zeros

Proof of Proposition 3(ii). Let s_n^* be optimal with $z_n = N_0(s_n^*)$. By Theorem 2 and Lemma 21,

$$\frac{n^2}{2} - 15n^{3/2} \leq a(n) = f(s_n^*) \leq \frac{n(n+1)}{2} - \frac{z_n(z_n+1)}{2}.$$

Hence

$$\frac{z_n(z_n+1)}{2} \leq \frac{n(n+1)}{2} - \frac{n^2}{2} + 15n^{3/2} = \frac{n}{2} + 15n^{3/2}.$$

Multiplying through by 2 and using $z_n(z_n+1) \geq z_n^2$:

$$z_n^2 \leq z_n(z_n+1) \leq n + 30n^{3/2} \leq 31n^{3/2} \quad \text{for all } n \geq 1.$$

Taking square roots, $z_n \leq \sqrt{31} n^{3/4}$, so $z_n = O(n^{3/4})$. Since $N_1(s_n^*) = n - z_n$,

$$\frac{N_1(s_n^*)}{n} = 1 - \frac{z_n}{n} \geq 1 - \frac{\sqrt{31}}{n^{1/4}} = 1 - O(n^{-1/4}). \quad \square$$

9 Upper bounds via the zero structure

Lemma 21 treats all zeros equally. We now refine the bound using the *run structure* of the zeros.

Let the maximal zero-runs of s have lengths $c_1, \dots, c_m$, with the i -th run beginning at position a_i and $r_i = n - (a_i + c_i) + 1$ positions to its right.

Lemma 22 (Run-sensitive bound). $f(s) \leq n(n+1)/2 - \sum_{i=1}^m (c_i r_i + \binom{c_i+1}{2})$.

Proof. Apply Lemma 21 and group by zero-runs. The i -th run occupies positions $a_i, \dots, a_i + c_i - 1$, contributing $\sum_{t=0}^{c_i-1} (n - a_i - t + 1) = c_i(n - a_i + 1) - c_i(c_i - 1)/2 = c_i r_i + \binom{c_i+1}{2}$ to the sum. $\square$

10 Fragmentation penalty

Corollary 23 (Fragmentation). *If s has z zeros in m maximal zero-runs, then*

$$f(s) \leq \frac{n(n+1)}{2} - \frac{z(z+1)}{2} - \binom{m}{2}.$$

Proof. It suffices to show $\sum_i (c_i r_i + \binom{c_i+1}{2}) \geq z(z+1)/2 + \binom{m}{2}$. Since consecutive zero-runs are separated by at least one 1, for $i < m$ we have $r_i \geq \sum_{j>i} c_j + (m - i)$, so $\sum_i c_i r_i \geq \sum_{i<j} c_i c_j + \sum_{i=1}^{m-1} c_i(m - i)$. The identity $\binom{z+1}{2} = \sum_{i<j} c_i c_j + \sum_i (c_i^2 + c_i)/2$ gives $\sum_{i<j} c_i c_j + \sum_i \binom{c_i+1}{2} = z(z+1)/2$. Finally $\sum_{i=1}^{m-1} c_i(m - i) \geq \sum_{i=1}^{m-1} (m - i) = \binom{m}{2}$ (each $c_i \geq 1$). $\square$

Remark 24. Corollary 23 shows that fragmenting the zeros into more runs always reduces the upper bound on $f(s)$, regardless of positions. Each additional run incurs an extra penalty of at least $m - 1$ (the new run number), on top of the baseline $z(z+1)/2$.

11 Convexity of the sequence

The computational data (Table 1 and [2]) show that the second differences $\Delta^2 a(n) := a(n+1) - 2a(n) + a(n-1)$ take only the values $\{0, 1, 2\}$ for all $n \leq 200$. We now prove one half of this observation: $\Delta^2 a(n) \geq 0$ for all $n \geq 2$. The proof is purely combinatorial and does not rely on the explicit construction of Section 4.

11.1 Background and strategy

Write $\Delta a(n) = a(n) - a(n-1)$ for $n \geq 2$. The claim $\Delta^2 a(n) \geq 0$ is equivalent to

$$a(n+1) \geq a(n) + \Delta a(n) = 2a(n) - a(n-1). \quad (19)$$

That is, we must exhibit a string of length $n+1$ achieving at least $a(n) + \Delta a(n)$ distinct substring values.

The strategy is to start from an *optimal* string s of length n and extend it by one bit (appending or prepending). The key insight is that the values in $V(s)$ that are “new” relative to a length- $(n-1)$ truncation of s can each be “lifted” to a genuinely new value in the extended string, and there are at least $\Delta a(n)$ such values.

11.2 Notation and setup

Fix $n \geq 2$ and let $s = s_1 s_2 \cdots s_n$ be an optimal string of length n , so $f(s) = a(n)$. Define the prefix and suffix of length $n-1$:

$$p = s_1 s_2 \cdots s_{n-1}, \quad q = s_2 s_3 \cdots s_n. \quad (20)$$

Since every substring of p (resp. q) is also a substring of s , we have $V(p) \subseteq V(s)$ and $V(q) \subseteq V(s)$.

Since p has length $n-1$, we have $f(p) \leq a(n-1)$. Therefore

$$|V(s) \setminus V(p)| = f(s) - f(p) \geq a(n) - a(n-1) = \Delta a(n), \quad (21)$$

and by the same argument applied to the suffix,

$$|V(s) \setminus V(q)| \geq \Delta a(n). \quad (22)$$

Each value $x \in V(s) \setminus V(p)$ is represented in s but not in the prefix p ; every witness for x must “touch” the last position. Symmetrically, each value in $V(s) \setminus V(q)$ has every witness touching the first position.

11.3 Unique standard representation

Recall that every positive integer x has a *standard binary representation*: the unique binary string starting with 1 and having value x . Its length is $\lfloor \log_2 x \rfloor + 1$. If w is any other binary string with $\nu(w) = x > 0$, then either w starts with 0 (leading zeros) or has the same length as the standard form; in the latter case it is the standard form itself.

For a positive $x \in V(s)$, define the *canonical witness* w_x to be a minimum-length substring of s with $\nu(w_x) = x$. Then:

Lemma 25. *If $x > 0$ then w_x is the standard binary representation of x .*

Proof. If w_x started with 0, removing the leading zero gives a strictly shorter substring of s with the same value, contradicting the minimality of w_x . So w_x starts with 1. Its length is at least $\lfloor \log_2 x \rfloor + 1$ (the length of the standard form); by minimality, equality holds, so w_x has the same length as the standard form. Since both start with 1 and represent x , they are the same binary string. $\square$

Proof of Theorem 4. Step 1: new positive values must touch the boundary.

Let $x \in V(s) \setminus V(p)$ with $x > 0$. Its canonical witness w_x is, by Lemma 25, the standard representation of x ; in particular w_x starts with 1.

Claim: w_x ends at position n (the last position of s).

Proof. Write $w_x = s_i \cdots s_j$. If $j \leq n-1$ then w_x is a substring of $p = s_1 \cdots s_{n-1}$, giving $x \in V(p)$ and contradicting $x \in V(s) \setminus V(p)$. Hence $j = n$.

Step 2: appending a bit creates a new value.

Fix any $b \in \{0, 1\}$ and form the extended string $sb = s_1 \cdots s_n b$ of length $n + 1$.

The word $w_x b$ occurs as a substring of sb ending at the new last position. Since w_x starts with 1 and has length $\ell := |w_x|$, the string $w_x b$ starts with 1 and has length $\ell + 1$, representing the integer $\nu(w_x b) = 2\nu(w_x) + b = 2x + b$.

Claim: $w_x b$ is the standard representation of $2x + b$.

Proof. Since w_x is the standard representation of x , we have $2^{\ell-1} \leq x < 2^\ell$, so $2^\ell \leq 2x + b < 2^{\ell+1}$. Hence the standard representation of $2x + b$ has exactly $\ell + 1$ bits. The string $w_x b$ starts with 1, has $\ell + 1$ bits, and has value $2x + b$; it is therefore the standard representation.

Claim: $2x + b \notin V(s)$.

Proof. Suppose $2x + b \in V(s)$. Take a minimum-length representative u of $2x + b$ in s . Since $2x + b > 0$, by Lemma 25 the substring u is the standard representation of $2x + b$, which we have just shown to be $w_x b$. Hence $u = w_x b$, and s contains $w_x b$ as a contiguous substring. Let $w_x b = s_i \cdots s_{i+\ell}$. Deleting the last character gives $w_x = s_i \cdots s_{i+\ell-1}$ with $i + \ell - 1 \leq n - 1$ (since $i + \ell \leq n$), so w_x is a substring of the prefix p . This gives $x = \nu(w_x) \in V(p)$, contradicting $x \in V(s) \setminus V(p)$.

Therefore $2x + b \in V(sb) \setminus V(s)$ for each positive $x \in V(s) \setminus V(p)$. Distinct values $x \neq x'$ yield distinct $2x + b \neq 2x' + b$ (the map $x \mapsto 2x + b$ is injective on the integers), so

$$f(sb) - f(s) \geq |\{x \in V(s) \setminus V(p) : x > 0\}| = |V(s) \setminus V(p)| - \mathbf{1}[0 \in V(s) \setminus V(p)], \quad (23)$$

where $\mathbf{1}[\cdot]$ denotes the indicator.

Step 3: the generic case ($0 \notin V(s) \setminus V(p)$).

If $0 \notin V(s) \setminus V(p)$, then every element of $V(s) \setminus V(p)$ is positive, and (23) becomes

$$f(sb) - f(s) \geq |V(s) \setminus V(p)| \geq \Delta a(n) \quad (\text{by (21)}).$$

Hence $a(n + 1) \geq f(sb) \geq a(n) + \Delta a(n)$, i.e. $\Delta a(n + 1) \geq \Delta a(n)$.

Step 4: the exceptional case ($0 \in V(s) \setminus V(p)$).

Assume $0 \in V(s) \setminus V(p)$. We show this forces a very specific structure and then switch to the suffix argument.

Since $0 \in V(s)$, the string s contains at least one 0. Since $0 \notin V(p)$, the prefix $p = s_1 \cdots s_{n-1}$ contains no zero, so $p = 1^{n-1}$. Combined with the existence of a zero in s and $s_1 = \cdots = s_{n-1} = 1$, we must have $s_n = 0$, so

$$s = 1^{n-1}0. \quad (24)$$

The suffix is $q = s_2 \cdots s_n = 1^{n-2}0$, which contains a zero, so $0 \in V(q)$. Therefore

$$0 \notin V(s) \setminus V(q), \quad (25)$$

and every element of $V(s) \setminus V(q)$ is positive.

Step 5: prepending a 1 creates a new value.

Let $x \in V(s) \setminus V(q)$. By (25), $x > 0$. Its canonical witness w_x is the standard representation of x (Lemma 25).

Claim: w_x starts at position 1 (the first position of s).

Proof. Write $w_x = s_i \cdots s_j$. If $i \geq 2$ then w_x is a substring of $q = s_2 \cdots s_n$, giving $x \in V(q)$, contradicting $x \in V(s) \setminus V(q)$. Hence $i = 1$.

Form the extended string $1s = 1s_1 \cdots s_n$ of length $n + 1$. Since w_x starts at position 1 of s , the word $1w_x$ occurs as a prefix of $1s$ (starting at position 1 of $1s$).

Let $\ell = |w_x|$ and define $y = \nu(1w_x) = 2^\ell + x$.

Claim: $1w_x$ is the standard representation of y .

Proof. Since w_x is the standard representation of x with ℓ bits, we have $2^{\ell-1} \leq x < 2^\ell$, so $2^\ell \leq y = 2^\ell + x < 2^{\ell+1}$. Hence the standard representation of y has exactly $\ell + 1$ bits. The string $1 w_x$ starts with 1, has $\ell + 1$ bits, and has value y ; it is therefore the standard representation.

Claim: $y \notin V(s)$.

Proof. Suppose $y \in V(s)$. Take a minimum-length representative u of y in s . Since $y > 0$, by Lemma 25 u is the standard representation of y , namely $1 w_x$. So s contains $1 w_x$ as a contiguous substring, say $1 w_x = s_i \cdots s_{i+\ell}$. Deleting the first character gives $w_x = s_{i+1} \cdots s_{i+\ell}$ with $i + 1 \geq 2$, so w_x is a substring of $q = s_2 \cdots s_n$. This gives $x \in V(q)$, contradicting $x \in V(s) \setminus V(q)$.

Therefore $y \in V(1s) \setminus V(s)$ for each $x \in V(s) \setminus V(q)$.

Claim: distinct values of x yield distinct values of y .

Proof. Let $x \neq x'$ with canonical witnesses of lengths ℓ and ℓ' respectively. Then $y = 2^\ell + x$ and $y' = 2^{\ell'} + x'$. If $\ell = \ell'$, then $y \neq y'$ since $x \neq x'$. If $\ell < \ell'$, then $y = 2^\ell + x < 2^{\ell+1} \leq 2^{\ell'} \leq y'$. In both cases $y \neq y'$.

Hence

$$f(1s) - f(s) \geq |V(s) \setminus V(q)| \geq \Delta a(n) \quad (\text{by (22)}),$$

and $a(n+1) \geq f(1s) \geq a(n) + \Delta a(n)$, i.e. $\Delta a(n+1) \geq \Delta a(n)$.

In both the generic case (Step 3) and the exceptional case (Steps 4–5), we have shown $\Delta a(n+1) \geq \Delta a(n)$, completing the proof. $\square$

Remark 26. Theorem 4 shows $\Delta^2 a(n) \geq 0$. Computational data [2] further suggest $\Delta^2 a(n) \leq 2$ for all $n \geq 2$: the second differences take only the values $\{0, 1, 2\}$ for all $n \leq 200$, with the value 2 occurring at seven values of n (52, 71, 117, 157, 178, 191, 194). Proving the upper bound $\Delta^2 a(n) \leq 2$ remains open.

12 Computational evidence

Known exact values ($n \leq 80$). Exact values for $n \leq 80$ appear in the OEIS b-file submitted by M. Fuller in 2008/9 [1]. No algorithm or proof of completeness was published; the values have the status of well-verified computational claims.

Heuristic values ($81 \leq n \leq 150$). Values of $a(n)$ for $81 \leq n \leq 150$ were computed in [2] using three independent heuristic methods: unconstrained Metropolis–Hastings search (at most 10 restarts), constrained Metropolis–Hastings with a fixed run-length template, and exhaustive template search over empirically derived block-range constraints. All three methods agreed on every overlapping case, giving high confidence in the values; however, these results are *not formally proved to be exact* and should be understood as heuristic computational estimates.

Large-scale certified lower bounds. A suffix-array-guided local search (*surgical nudge*) was used for n up to 2×10^9 [2]. In each iteration the algorithm finds the highest-LCP collision in the suffix array, searches a neighbourhood for a density-preserving swap that resolves it, and accepts only swaps that strictly increase the exact score. Since the score is monotone non-decreasing, every recorded value is a rigorous lower bound. At $n = 2 \times 10^9$ a lower bound of $a(n)/n^2 \geq 0.49996$ was obtained.

Comparison with theory. The ratio $a(n)/n^2$ rises from 0.3596 at $n = 50$ to 0.4031 at $n = 150$, and exceeds 0.49996 at $n = 2 \times 10^9$. Theorem 2 gives $a(n)/n^2 \geq 1/2 - 15/\sqrt{n}$, which reaches 0.4875 only at $n = 14,400$. The computational lower bounds substantially exceed the theoretical bound at moderate n , confirming that the constant 15 in $15n^{3/2}$ is far from tight.

Entropy of optimal solutions. Let S_n denote the set of all strings achieving $a(n)$ and $H(n) = \log_2 |S_n|$. For $n \leq 150$, $H(n)$ is almost always zero or very small: the typical optimum is unique or nearly so, with $H(n) \leq 10.4$ throughout. Optimal strings share long common prefixes; the zero-separator structure is completely rigid in 28 cases and varies by at most one separator in a further 119 cases out of $n \leq 150$.

13 Heuristic remarks

To maximise $f(s)$, one seeks to maximise the number of leading-1 substrings that remain pairwise distinct after leading zeros are stripped. Long repeated patterns create large common-prefix collisions, inflating LCP values and causing many pairs of substrings to agree. The explicit construction in Section 4 avoids this by using *strictly decreasing* one-block lengths: any two substrings $u_{i,j,\alpha,\beta}$ and $u_{i',j',\alpha',\beta'}$ spanning different block-pairs $(i,j) \neq (i',j')$ have different interior block-length sequences, making them provably distinct (Lemma 8).

Empirically, all optimal strings for $n \leq 150$ follow a similar principle: they begin with a long run of ones, followed by decreasing-length one-blocks with a fixed separator prefix $1, 2, 1, 1, \dots$, and have ones-density growing toward 1. This structure is consistent with the proven density bounds: the empirical zero count at small n is $O(\sqrt{n})$, well within the theoretical upper bound of $O(n^{3/4})$ and matching the zero count of the explicit construction in Section 4.

Acknowledgements

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